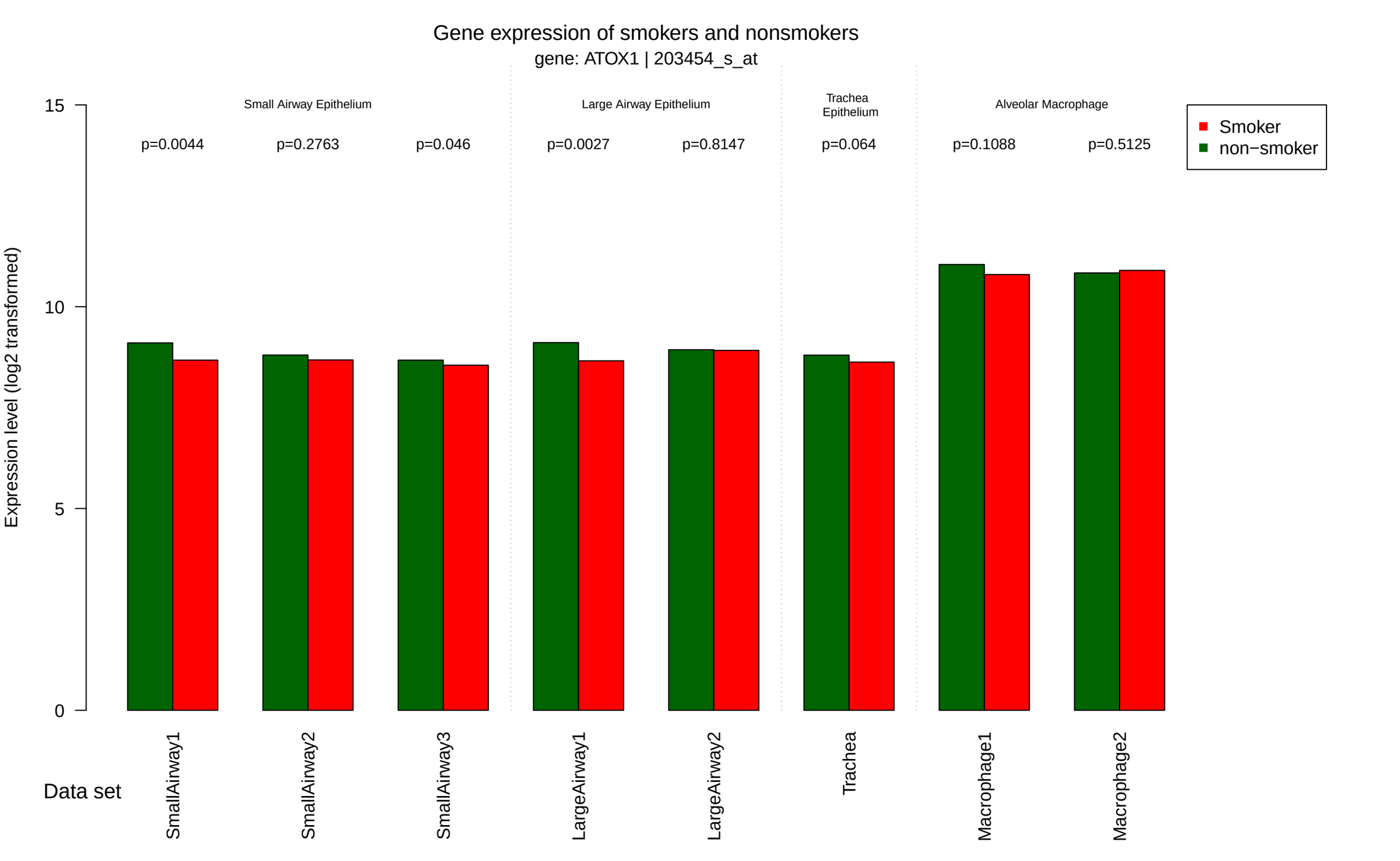
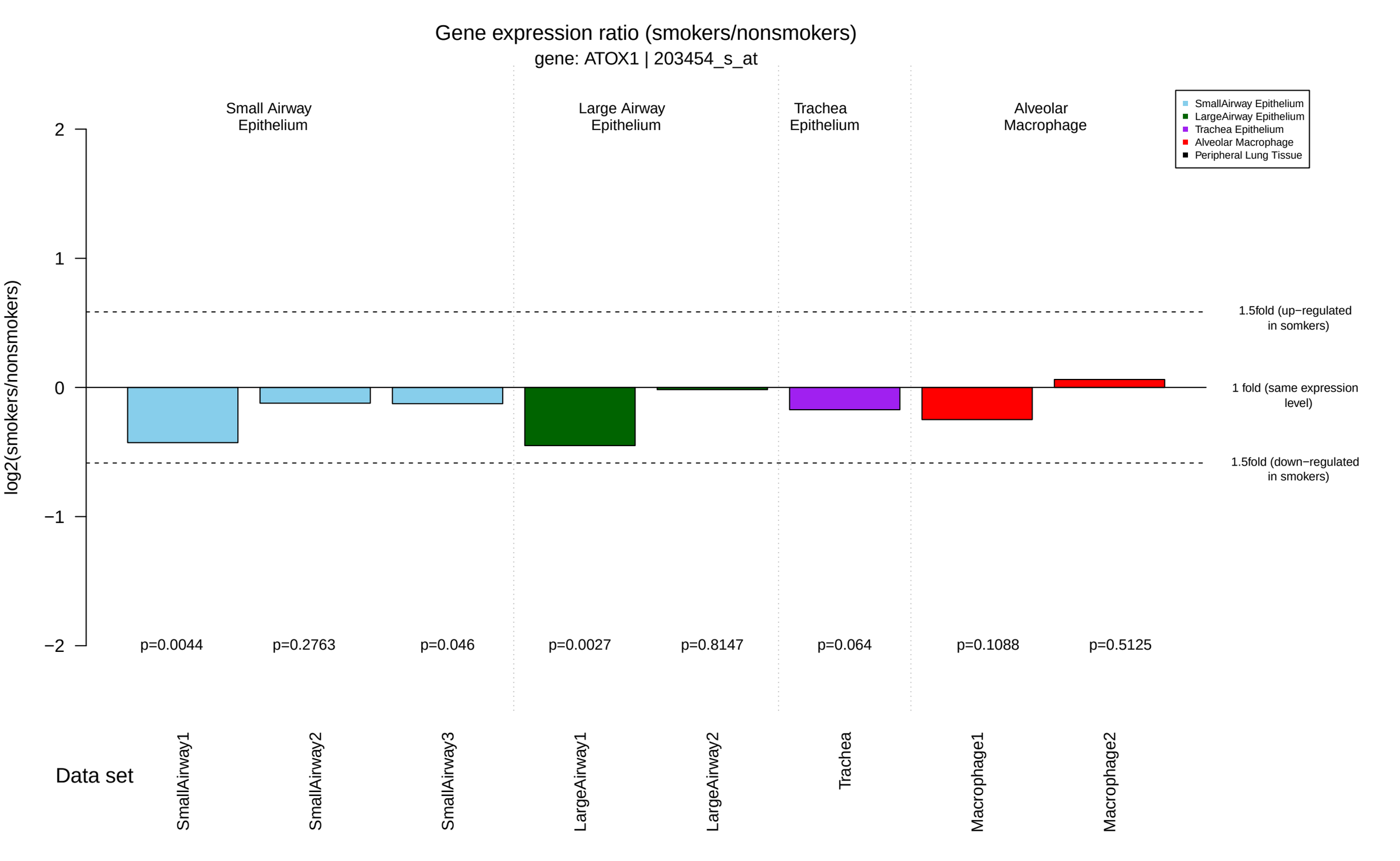
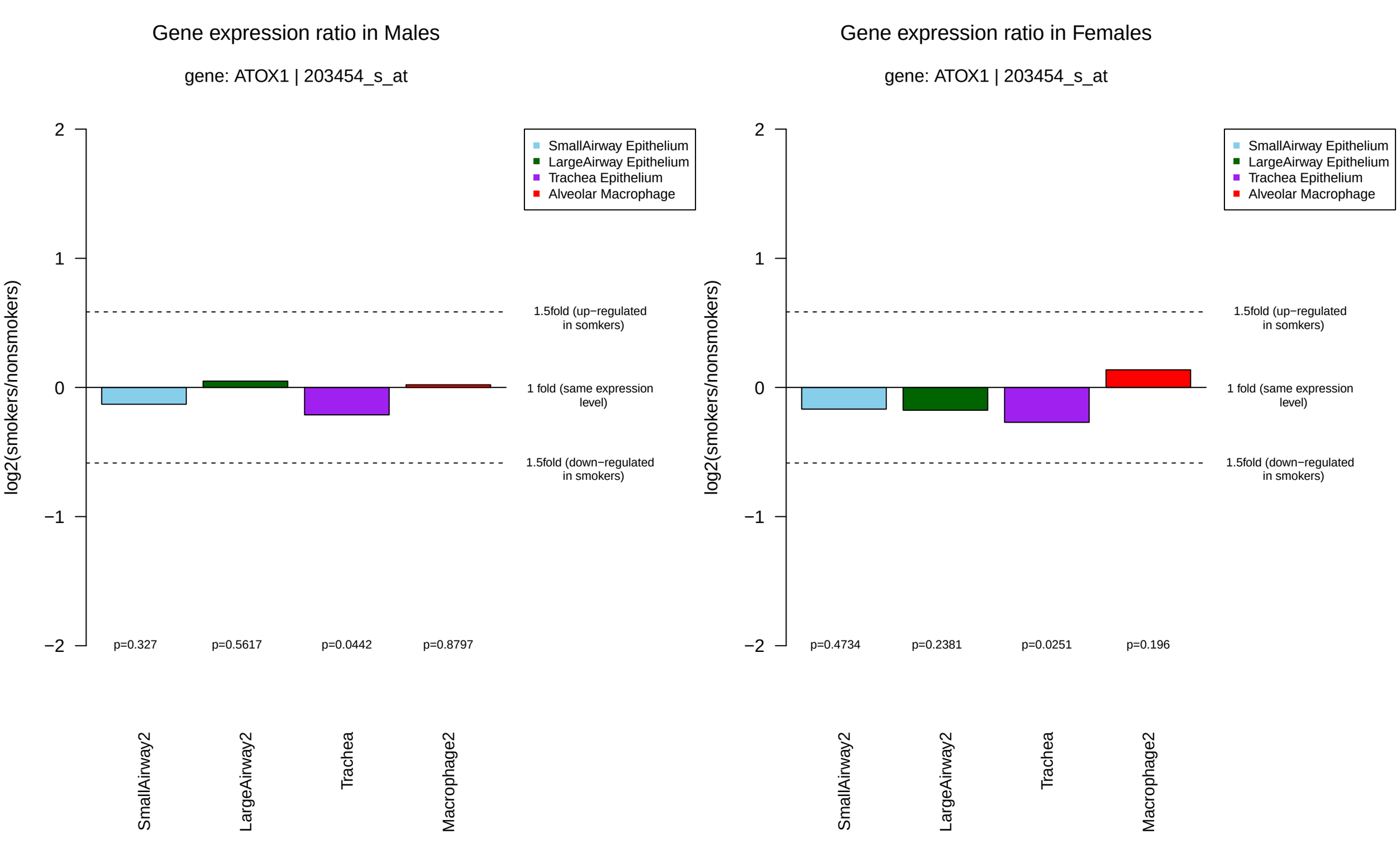
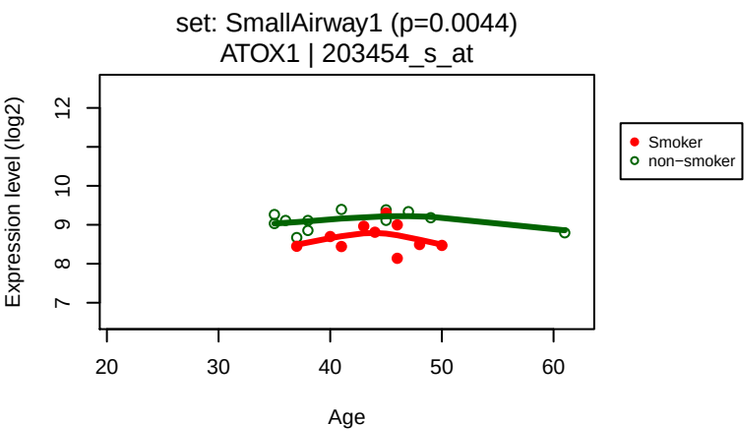
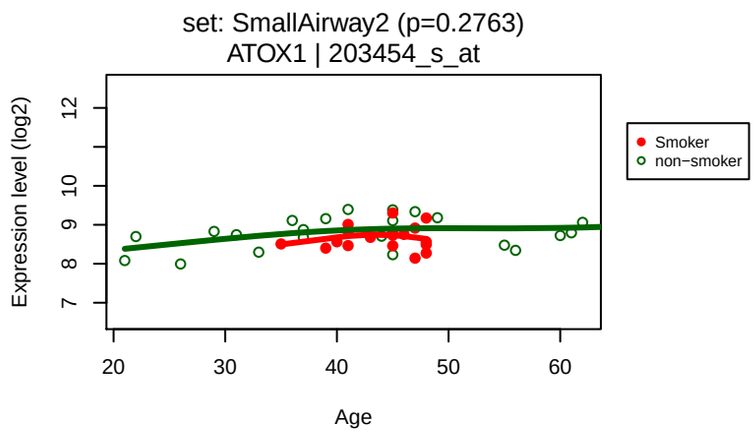
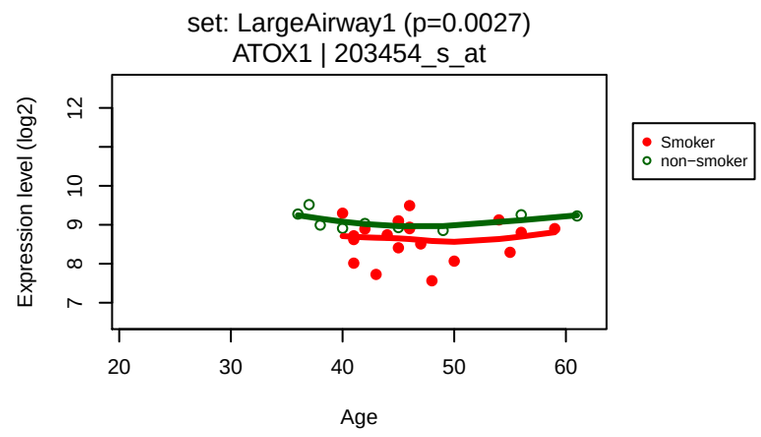

Gene expression of smokers and nonsmokers
gene: ATOX1 | 203454_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0044
p=0.2763
p=0.046
p=0.0027
p=0.8147
p=0.064
p=0.1088
p=0.5125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

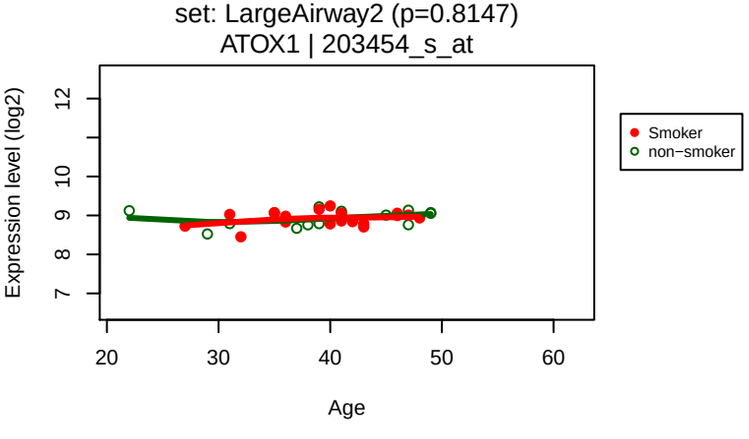

Gene expression ratio (smokers/nonsmokers)
gene: ATOX1 | 203454_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0044
p=0.2763
p=0.046
p=0.0027
p=0.8147
p=0.064
p=0.1088
p=0.5125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

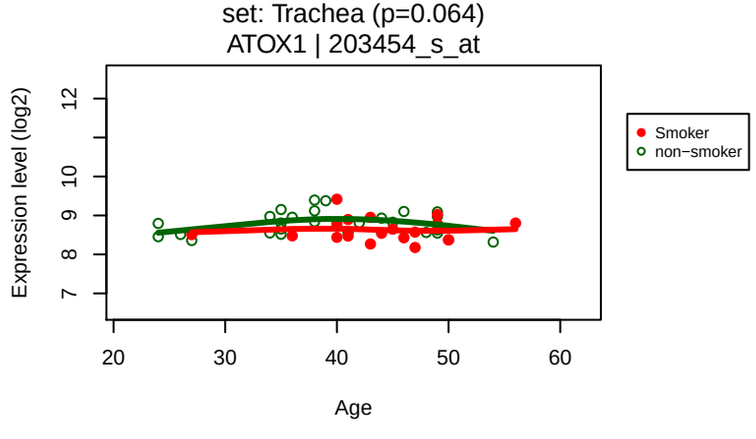

Gene expression ratio in Males
gene: ATOX1 | 203454_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.327
p=0.5617
p=0.0442
p=0.8797
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATOX1 | 203454_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4734
p=0.2381
p=0.0251
p=0.196
SmallAirway2
LargeAirway2
Trachea
Macrophage2

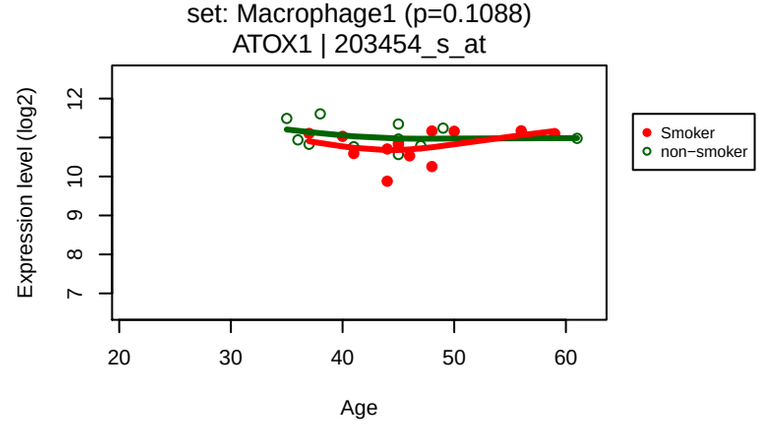

set: SmallAirway1 (p=0.0044)
ATOX1 | 203454_s_at
Expression level (log2)
Age
Smoker
non−smoker

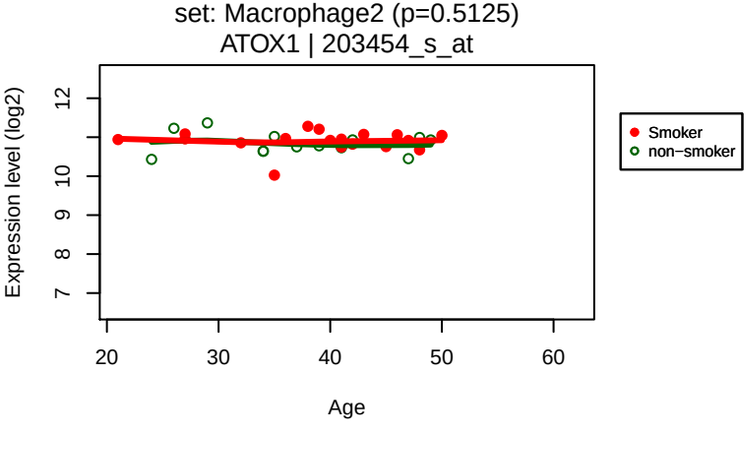

set: SmallAirway2 (p=0.2763)
ATOX1 | 203454_s_at
Expression level (log2)
Age
Smoker
non−smoker

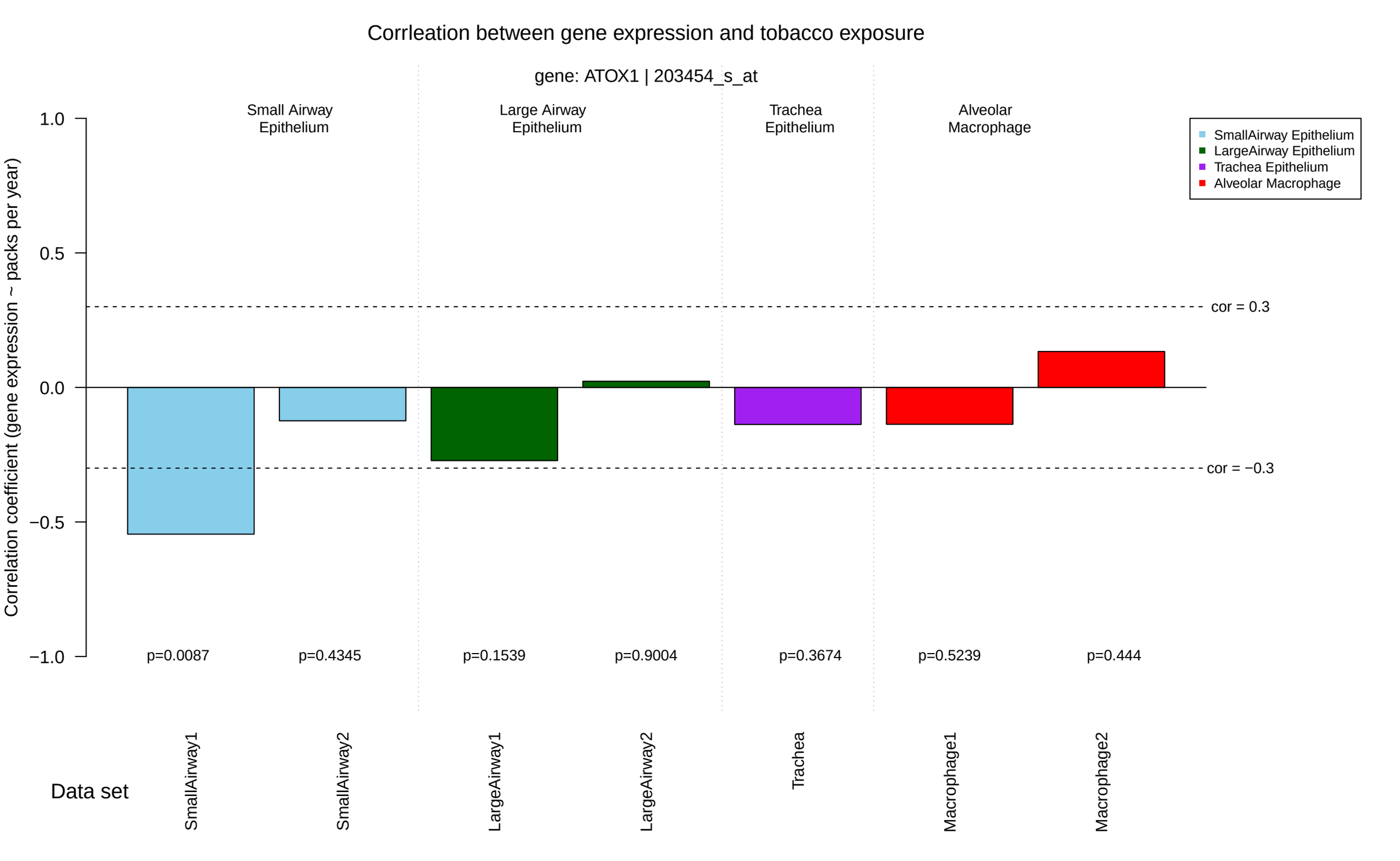

set: LargeAirway1 (p=0.0027)
ATOX1 | 203454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8147)
ATOX1 | 203454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.064)
ATOX1 | 203454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1088)
ATOX1 | 203454_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5125)
ATOX1 | 203454_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATOX1 | 203454_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0087
p=0.4345
p=0.1539
p=0.9004
p=0.3674
p=0.5239
p=0.444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2